

Lecture 14

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Lectures_
MA5104

MA 5104: Statistical Inference and Learning

Module 2: Estimation

Lecture 14: Bayesian Estimation and Uncertainty

Dr. Rakhi Singh

Department of Mathematics

IIT Madras

- Balance data (x) w/ prior beliefs

- Distribution for θ

- $\rightarrow$ uncertainty quantification

- $\rightarrow$ Bayesian update

- Sensitivity to prior

- Subjective prior

- computational complexity

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From estimating a parameter to quantifying uncertainty

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x | \theta).$$

After observing

$$X_1 = x_1, \dots, X_n = x_n,$$

we want to learn about θ .

Frequentist approach

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta; x)$$

$\text{SE}(\hat{\theta}),$ confidence interval

Bayesian approach

Represent uncertainty about θ using a probability distribution:

$$\pi(\theta | x)$$

What does the data tell us about the unknown parameter?

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Suppose you have some data coming in for outcomes of disease. You know at $D_0 \rightarrow 2$, $D_1 \rightarrow 2$, $D_2 \rightarrow 2$, $D_4 \rightarrow 10$, ... can you predict if you have sufficient beds?

$\rightarrow$ Reinforcement learning?

The Bayesian model

We specify a probability model for both the data and the unknown parameter:

$\theta \sim \pi(\theta)$

$X | \theta \sim f(x | \theta)$

After observing x ,

Posterior on θ

$$\pi(\theta | x) = \frac{f(x | \theta)\pi(\theta)}{\int f(x | t)\pi(t) dt}$$

Bayes thm

Hence,

$$\pi(\theta | x) \propto L(\theta; x)\pi(\theta)$$

Posterior
updated uncertainty

$\propto$

Likelihood
information from data

$\times$

Prior
information before data

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advantages

- Bayesian updates
- Embed information about prior
- Uncertainty quantification

disadvantages

- computational complexity
- no closed form soln often
- sensitive to prior

Likelihood and posterior are different objects

For observed data x , $L(\theta; x) = f(x | \theta)$ is a function of θ .

The posterior is

$$\pi(\theta | x) \propto L(\theta; x)\pi(\theta).$$

Likelihood

Measures how compatible different values of θ are with the observed data.

$$L(\theta; x)$$

Posterior

Represents uncertainty about θ after observing the data.

$$\pi(\theta | x)$$

$$\text{Posterior} = \text{Likelihood} \times \text{Prior, normalized}$$

Example: Bayesian estimation of a normal mean

Suppose $X_1, \dots, X_n | \mu \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, where σ^2 is known.

Assume

$$\mu \sim N(\mu_0, \tau^2).$$

Then

$$\mu | x \sim N(\mu_n, \tau_n^2),$$

where

$$\frac{1}{\tau_n^2} = \frac{1}{\tau^2} + \frac{n}{\sigma^2} \text{ and } \mu_n = \frac{\mu_0/\tau^2 + n\bar{X}/\sigma^2}{1/\tau^2 + n/\sigma^2}.$$

Equivalently,

$$\mu_n = w\mu_0 + (1-w)\bar{X} \text{ for } w = \frac{\sigma^2}{\sigma^2 + n\tau^2}.$$

What is the Bayesian point estimate?

Once we have the posterior distribution, there are several possible summaries.

Posterior mean

$$\hat{\theta}_{\text{PM}} = \mathbb{E}(\theta | x)$$

Posterior median

$$\hat{\theta}_{\text{Med}} = \text{median}(\theta | x)$$

Maximum a posteriori estimate

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \pi(\theta | x) \quad \text{MAP}$$

There is no universally preferred Bayesian point estimate.

The MAP–MLE connection

The MLE is $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta; x)$.

The MAP estimate is

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \pi(\theta | x).$$

Since

$$\pi(\theta | x) \propto L(\theta; x)\pi(\theta),$$

we have

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} L(\theta; x)\pi(\theta)$$

or, using log-likelihood,

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \{\ell(\theta; x) + \log \pi(\theta)\}.$$

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MLE as a special case of MAP

Suppose the prior is flat:

$$\pi(\theta) \propto 1. \quad \nexists \theta$$

Then

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} L(\theta; x)\pi(\theta) \text{ becomes } \hat{\theta}_{\text{MAP}} = \arg \max_{\theta} L(\theta; x).$$

Therefore,

$$\boxed{\text{MLE} = \text{MAP under a flat prior.}}$$

But: “flat” is not always innocuous.

The meaning of a flat prior can depend on the parameterization. In particular, MAP estimates are not invariant under nonlinear reparameterizations.

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A prior can act like a penalty

Consider $\theta \sim \mathcal{N}(0, \tau^2)$. Then

$$\log \pi(\theta) = -\frac{\theta^2}{2\tau^2} + C.$$

Therefore,

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \left[\ell(\theta; x) - \frac{\theta^2}{2\tau^2} \right].$$

Equivalently,

$$\hat{\theta}_{\text{MAP}} = \arg \min_{\theta} \left[-\ell(\theta; x) + \lambda \theta^2 \right], \text{ where } \lambda = \frac{1}{2\tau^2}.$$

$$\boxed{\text{Gaussian prior} \longleftrightarrow \text{quadratic regularization}}$$

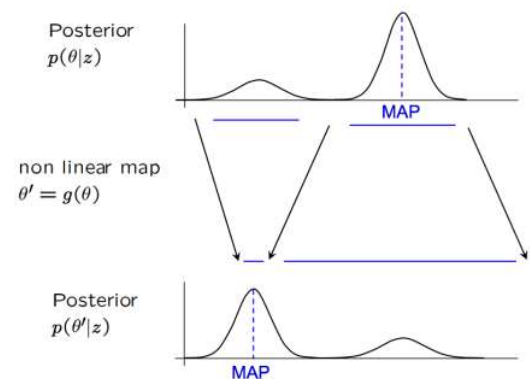
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Warning: MAP and change of parameterization



Problem:

- MAP finds maximum value of the posterior, but the posterior is a probability density

Solution:

- Loss functions correctly transform under change of coordinate (reparameterization)

The posterior is more than a point estimate

The Bayesian analysis does not need to end with $\hat{\theta}$.

Instead, we retain the entire posterior: $\pi(\theta | x)$ which allows us to ask:

- Where is θ most likely? $\rightarrow$ MAP $\checkmark$
- What is the posterior center? $\rightarrow$ posterior mean/median $\leftarrow$
- How uncertain are we about θ ? $\rightarrow$ posterior spread $\leftarrow$
- Which values of θ are plausible? $\rightarrow$ credible interval $\leftarrow$

The posterior distribution itself is the inferential object.

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Bayesian credible intervals - not unique

A $100(1 - \alpha)\%$ credible interval is an interval $[L, U]$ satisfying

$$\Pr_{\theta}(L \leq \theta \leq U | x) = 1 - \alpha.$$

For example, a 95% credible interval satisfies

$$\Pr_{\theta}(L \leq \theta \leq U | x) = 0.95.$$

If $\theta | x \sim \mathcal{N}(\mu_n, \tau_n^2)$, then the central 95% credible interval is

$$\mu_n \pm 1.96\tau_n.$$

The probability statement is about θ given the observed data.

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Credible interval versus confidence interval

Suppose the resulting interval is $[2.1, 3.4]$.

Bayesian

$$\Pr_{\theta}(2.1 \leq \theta \leq 3.4 | x) = 0.95.$$

Given the observed data and prior, there is 95% posterior probability that θ lies in the interval.

Frequentist

$$\Pr_{\theta}\{\theta \in C(X)\} = 0.95.$$

The procedure has 95% coverage over repeated samples.

credible interval: probability about parameter

confidence interval: probability about procedure

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Two ways of thinking about uncertainty

Frequentist

The parameter θ is fixed.
The estimator is random:

Bayesian

The observed data x are fixed.
Uncertainty about θ is represented by

The parameter θ is fixed.
The estimator is random:

$$\hat{\theta} = \hat{\theta}(X).$$

Uncertainty is described through the sampling distribution

$$p(\hat{\theta} | \theta).$$

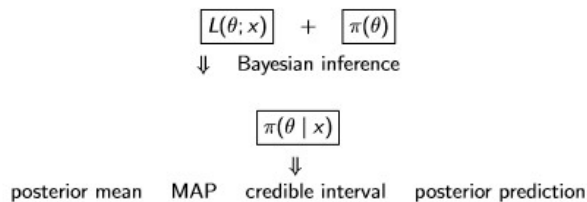
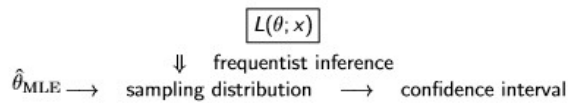
The observed data x are fixed.
Uncertainty about θ is represented by

$$\pi(\theta | x).$$

Inference is based on the posterior distribution.

The two approaches attach probability to different objects.

The bigger picture



A question for the next lecture

We have seen that frequentist uncertainty is connected to the sampling distribution of an estimator:

$$\hat{\theta} = T(X_1, \dots, X_n).$$

But what if the sampling distribution of $\hat{\theta}$ is difficult to derive?

Can we use the data themselves to learn about the sampling variability of $\hat{\theta}$?

Jackknife $\rightarrow$ Bootstrap